THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION MIL-N-25027

DD Form 672, MAY 88 PREVIOUS EDITIONS ARE OBSOLETE

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY-AV NAVY-AS
AIR FORCE-11 DLA-
REVIEW 82
USER CR, MI
PROJECT NUMBER 5310-1946
DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

MILITARY SPECIFICATION SHEET

TITLE NUT, SELF-LOCKING PLATE, ONE LUG LOW HEIGHT, STEEL, 125 KSI Ftu 450F

AMSC- N/A FSC 5310

SPECIFICATION SHEET NUMBER MS21051 1 FEB 94 REV G

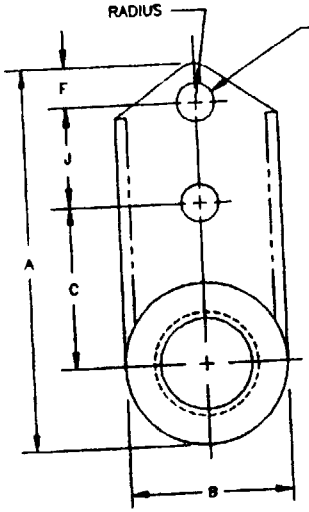
SUPERSEDING MS21051 F 30 SEP 87 (SEE NOTE 5.)

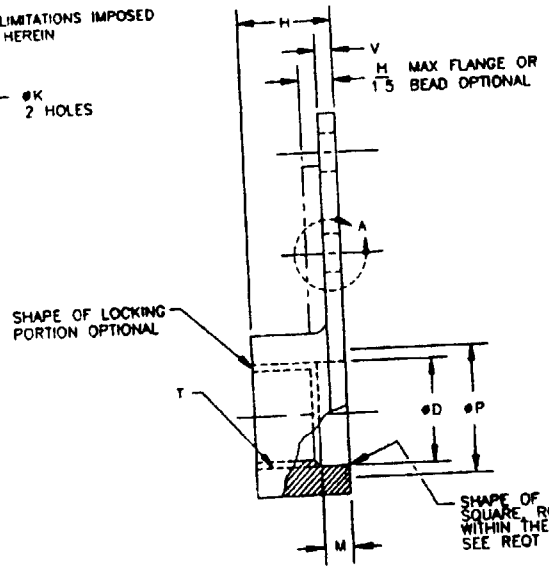
Page 1 of 2

Ⓖ DENOTES CHANGE(S)

INCH-POUND

CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS SPECIFIED HEREIN





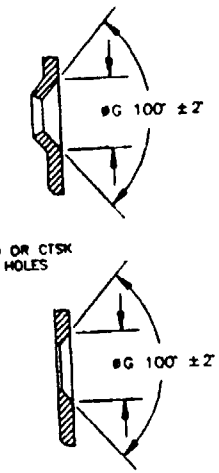


TABLE I DASH NUMBERS AND DIMENSIONS

DASH NUMBERS				T THREAD SIZES	A MAX	B		C ± 005	ØD MIN	F MIN	ØG ± 010	1/ H MAX	2/ J ± 002	3/ ØK + 005 - 000	M MIN	ØP MAX	V MAX	AXIAL STRENGTH LBS/MIN	WEIGHT LBS/100
PLAIN RIVET HOLES		COUNTERSUNK OR DIMPLED RIVET HOLES				MAX	MIN												
NON-DRY FILM LUBRICANT	DRY FILM LUBRICANT	NON-DRY FILM LUBRICANT	DRY FILM LUBRICANT																
-06	L06	-06K	L06K	138-32 UNJC-3B	935	297	239	344	—	100	200	171	312	098	—	208	047	1,130	28
-08	L08	-08K	L08K	164-32 UNJC-3B	935	297	277	344	344	100	200	250	312	098	—	248	047	1,720	29
-3	L3	-3K	L3K	190-32 UNJF-3B	950	328	308	344	194	100	200	250	312	098	082	274	047	2 460	33
-4	L4	-4K	L4K	250-28 UNJF-3B	993	414	375	344	254	100	200	281	312	098	082	344	055	4,580	57
-5	L5	-5K	L5K	3125-24 UNJF-3B	1 210	505	485	500	317	125	230	328	312	130	082	417	065	7 390	96
-6	L6	—	—	375-24 UNJF-3B	1 264	614	594	500	379	125	230	344	312	130	082	505	075	11 450	1 44
-7	L7	—	—	4375-20 UNJF-3B	1 446	726	697	562	442	156	230	390	344	161	082	602	085	15 450	2 02

1/ MINIMUM "H" NOT SPECIFIED, LIMITED ONLY BY STRENGTH REQUIREMENTS OF THE SPECIFICATION
2/ CENTER OF THE TAPPED HOLE SHALL NOT DEVIATE IN ANY DIRECTION FROM THE CENTER OF THE PLATE NOT AS DETERMINED BY THE RIVET HOLES BY MORE THAN 005
3/ DIMPLED RIVET HOLE TOLERANCE FOR "ØK" IS + 015 - 000

Form Approved OMB No 0704-0188

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:25Z
Check the source to verify that this is the current version before use.

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION MIL-N-25027

Form Approved
OMB No 0704-0188

REQUIREMENTS

(G) 1 MATERIAL CARBON STEEL COMPOSITIONS 1035 (UNS G10350) 1040 (UNS G10400) AND 1050 (UNS G10500) IN ACCORDANCE WITH ASTM A827 OR QQ-S-700 1042 (UNS G10420) IN ACCORDANCE WITH ASTM A29 ALLOY STEEL GRADES 4130 (UNS G41300) 4340 (UNS G43400) AND 8740 (UNS G87400) IN ACCORDANCE WITH ASTM A29

2 FINISH CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416 TYPE II CLASS 2 FOR DRY FILM LUBRICATED NUTS THE TYPE AND CLASS ARE OPTIONAL IF THE NUTS MEET THE SALT SPRAY REQUIREMENTS OF QQ-P-416 TYPE II

3 DIMENSIONING AND TOLERANCING DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14 5M

4 HARDNESS 49HRC MAX

5 THREADS THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-S-8879

6 SURFACE TEXTURE SURFACE TEXTURE UNLESS OTHERWISE SPECIFIED SHALL NOT EXCEED 125 MICROINCHES IN ACCORDANCE WITH ANSI/ASME B46 1

7 LUBRICANT DRY FILM LUBRICANT APPROVED IN ACCORDANCE WITH MIL-N-25027 NON-DRY FILM LUBRICANTS SHALL BE SOLUBLE IN THE CLEANER SPECIFIED IN MIL-S-8802

(G) 8 EDGES BREAK ALL SHARP EDGES ANDS REMOVE ALL BURRS

9 COUNTERBORE/COUNTERSINK ON SIZE 164 AND LARGER COUNTERBORE THREAD "D" DIAMETER X 062 DEEP MINIMUM THREAD RELIEF ON SIZE 138 AND SMALLER COUNTERSINK TO "øP" WITHIN THE LIMITS OF D P AND M

10 PART NUMBER THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I

EXAMPLE MS21051L4

- L4 — DASH NUMBER
- MS21051 — BASIC MS NUMBER

MS21051L4 INDICATES NUT SELF-LOCKING PLATE ONE LUG LOW HEIGHT STEEL 125 KSI Ftu 450°F 250-28 UNJF-3B DRY FILM LUBRICANT COUNTERSUNK OR DIMPLED RIVET HOLES

NOTES

1 ALL DIMENSIONS ARE IN INCHES

2 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE

3 REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN

4 DESIGN AND USAGE LIMITATIONS THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588 ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED

5 MS21051 SUPERSEDES NAS682 NAS1025 IN PART

INTERCHANGEABILITY RELATIONSHIP

MS21051 NUTS CAN UNIVERSALLY REPLACE NAS682 NUTS OF LIKE MATERIAL THREAD SIZE LUBRICANT (DRY FILM OR NON-DRY FILM LUBRICANT) AND FASTENING METHOD (PLAIN RIVET HOLES DIMPLED OR COUNTERSUNK RIVET HOLES) BUT THESE NAS682 NUTS CANNOT UNIVERSALLY REPLACE MS21051 NUTS

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS
NAS682X06	MS21051-06
NAS682X06K	MS21051-06K
NAS682A06	MS21051L06
NAS682A06K	MS21051L06K
NAS682X08	MS21051-08
NAS682X08K	MS21051-08K
NAS682A08	MS21051L08
NAS682A08K	MS21051L08K
NAS682X3	MS21051-3
NAS682X3K	MS21051-3K
NAS682A3	MS21051L3
NAS682A3K	MS21051L3K
NAS682X4	MS21051-4
NAS682X4K	MS21051-4K
NAS682A4	MS21051L4
NAS682A4K	MS21051L4K
NAS682X5	MS21051-5
NAS682X5K	MS21051-5K
NAS682A5	MS21051L5
NAS682A5K	MS21051L5K
NAS682X6	MS21051-6
NAS682A6	MS21051L6
NAS682X7	MS21051-7
NAS682A7	MS21051L7

PREPARING ACTIVITY DLA-IS CUSTODIANS ARMY-AV NAVY-AS AIR FORCE-11 DLA- REVIEW 82 USER CR MI PROJECT NUMBER 5310-1946	MILITARY SPECIFICATION SHEET TITLE NUT, SELF-LOCKING PLATE ONE LUG LOW HEIGHT STEEL 125 KSI Ftu 450 F	SPECIFICATION SHEET NUMBER MS21051 1 FEB 94 REV G SUPERSEDING MS21051 F 30 SEP 87 (SEE NOTE 5) AMSC- N/A FSC 5310

DISTRIBUTION STATEMENT
A Approved for public release distribution is unlimited

DD Form 672. MAY 88 PREVIOUS EDITION ARE OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:25Z
Check the source to verify that this is the current version before use.